Appendix A

Appendix

A.1 Useful Math References

A.1.1 Trigonometric Identities

$$\sin^2 x + \cos^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$1 + \cot^2 x = \csc^2 x$$

$$\sin(2x) = 2\sin x \cos x$$
$$\cos(2x) = \cos^2 x - \sin^2 x$$
$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

$$\sin\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{2}}$$
$$\cos\frac{x}{2} = \pm\sqrt{\frac{1+\cos x}{2}}$$
$$\tan\frac{x}{2} = \frac{1-\cos x}{\sin x} = \frac{\sin x}{1+\cos x}$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$
$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$
$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

A.1.2 Taylor Series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\frac{1}{1+x} = 1 + x + x^2 + x^3 + \dots \{-1 < x < 1\}$$

A.2 Helpful Proofs

A.2.1 Arc length and a Straight Line

Consider an arc subtended by a very small angle $d\theta$ in a circle of radius R.

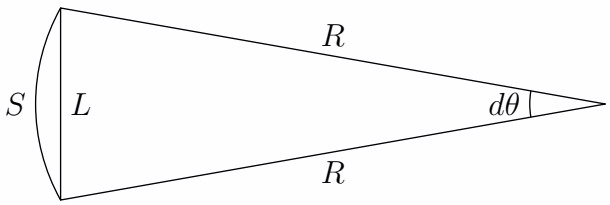

Fig. A.1

The distance S equals $R\theta$ while the distance L equals $2R\sin(d\theta/2)$. Taking the difference of these two lengths gives

$$\begin{aligned} S - L &= R\theta - 2R\sin\left(\frac{d\theta}{2}\right) \\ &= Rd\theta - 2R\left(\frac{d\theta}{2} - \frac{d\theta^3}{48} + \frac{d\theta^5}{3840} - \dots\right) \\ &\approx \frac{d\theta^3}{24} \end{aligned} \tag{A.1}$$

which is third-order small and thus will always be smaller than $d\theta$, which is already essentially zero. Since the two segments only differ to the third order in $d\theta$, we consider them to be equal for small angles ★

A.2.2 Properties of Ellipses

There is some important ellipse notation that should be understood before approaching elliptical orbits. Let us start with a diagram of an ellipse (note that it's not to scale):

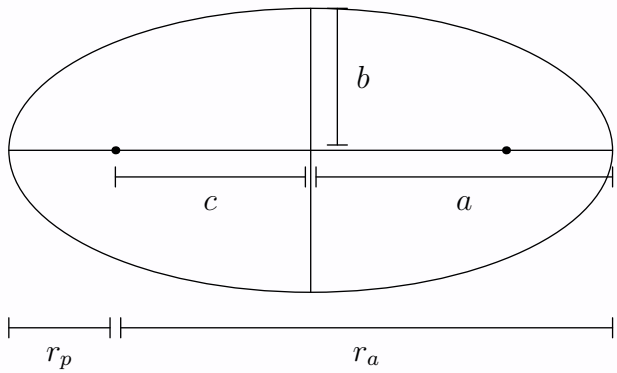

Fig. A.2

The distance a is the semi-major axis of the ellipse while the distance b is the semi-minor axis. The distance c is a representation of the eccentricity of the ellipse, i.e $c = \epsilon a$. If we place the star at the leftmost focus, r_p represents the perihelion distance while r_a represents the aphelion distance. From just these definitions, we can write

$$2a = r_a + r_p \tag{A.2}$$

$$2c = r_a - r_p \tag{A.3}$$

Next, by the definition of an ellipse, the distance from the top of the ellipse to each foci must be a. So, we can write

$$b^2 + c^2 = a^2 \tag{A.4}$$

Multiplying eq. A.4 through by 4 and substituting using A.2 and A.3 gives

$$4b^2 + r_a^2 - 2r_a r_p + r_p^2 = r_a^2 + 2r_a r_p + r_p^2$$

$$b = \sqrt{r_a r_p} \tag{A.5}$$

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A.2.3 Small Angle Approximations

For small angles, we often make a few approximations for trigonometric functions. $\sin(\theta)$ is approximated to θ, since writing out a taylor expansion yields

$$\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

Ignoring higher-order terms just yields $\sin\theta \approx \theta$. Doing the same for cosine yields

$$1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

Ignoring high-order terms gives $\cos\theta \approx 1 - \frac{\theta^2}{2}$ ★